

CWA SCIENCE CLASSES

Concept With Abhishek | Class 12 — Chemistry

Chapter 1: The Solid State

1. Solids have definite mass, volume & shape; classified as Crystalline (long range order) and Amorphous (short range/no order, pseudo solids).
2. Types of Crystalline Solids: Molecular, Ionic, Metallic, Covalent (network) solids — based on nature of constituent particles and bonding.
3. Unit Cell: Smallest repeating unit that shows the full symmetry of the crystal structure.
4. Types of Unit Cells: Primitive (particles only at corners) and Centred (Body-centred, Face-centred, End-centred).
5. Packing Efficiency: Simple cubic ~52.4%, BCC ~68%, FCC/HCP (close packed) ~74%.
6. Coordination Number: Number of nearest neighbours of a particle in the crystal lattice.
7. Point Defects: Stoichiometric (Schottky, Frenkel) and Non-stoichiometric (metal excess, metal deficiency) defects affect electrical & mechanical properties.